

Voronoi tessellation-based analysis of molecular structures

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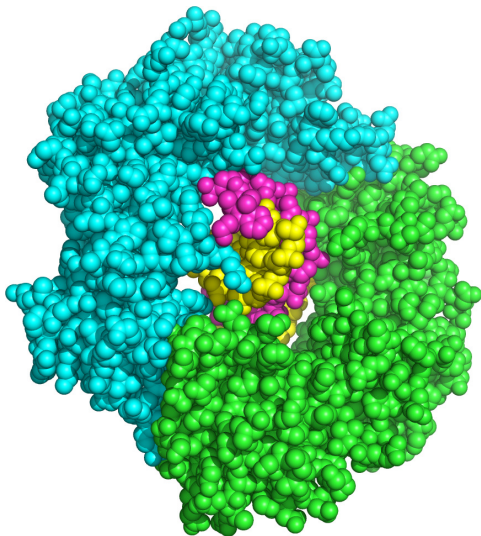
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When studying biological macromolecules, some common problems are:

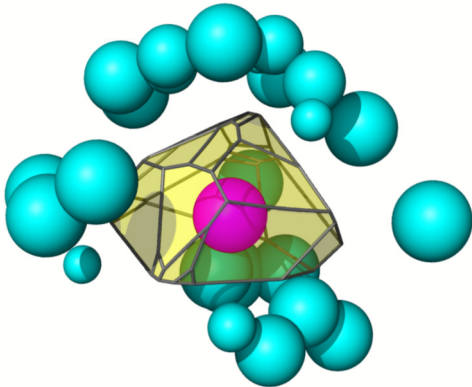
- ▶ analyzing how different parts in a molecule interact
- ▶ selecting the best prediction of a multimeric complex

Some possible solutions involve:

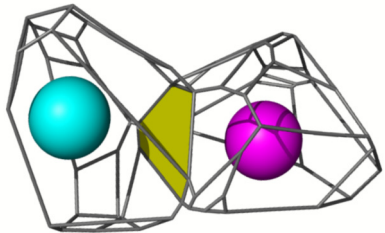
- ▶ computational geometry
- ▶ machine learning using open data
- ▶ developing free software

Deriving atom-atom contacts from the Voronoi tessellation

Voronoi cell of an atom surrounded by its neighbors

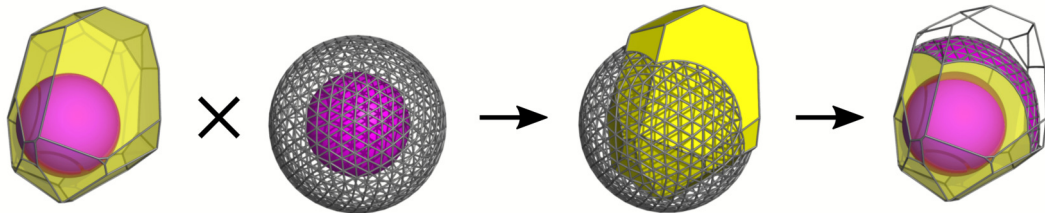


Atom-atom contact surface defined as the face shared by two adjacent Voronoi cells.



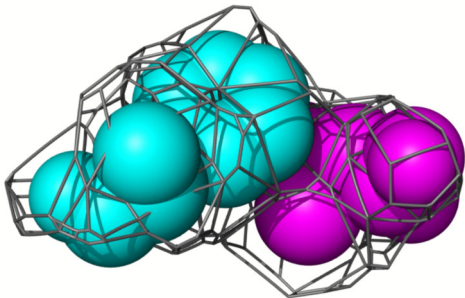
Constraining CSAs and deriving SASA for an atom

- ▶ **CSA** — Contact Surface Area
- ▶ **SASA** — Solvent-Accessible Surface Area

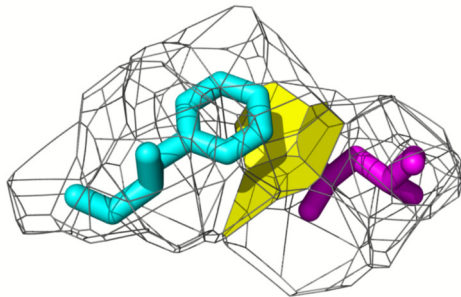


Deriving residue-residue contacts

Voronoi cells of two neighboring residues

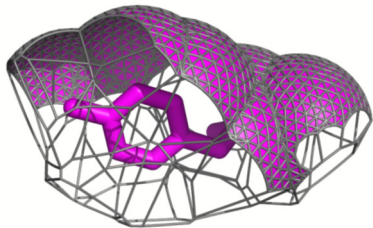


Residue-residue contact surface defined as a union of atom-atom contact surfaces

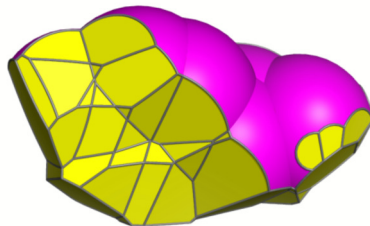


CSA and SASA of a single residue

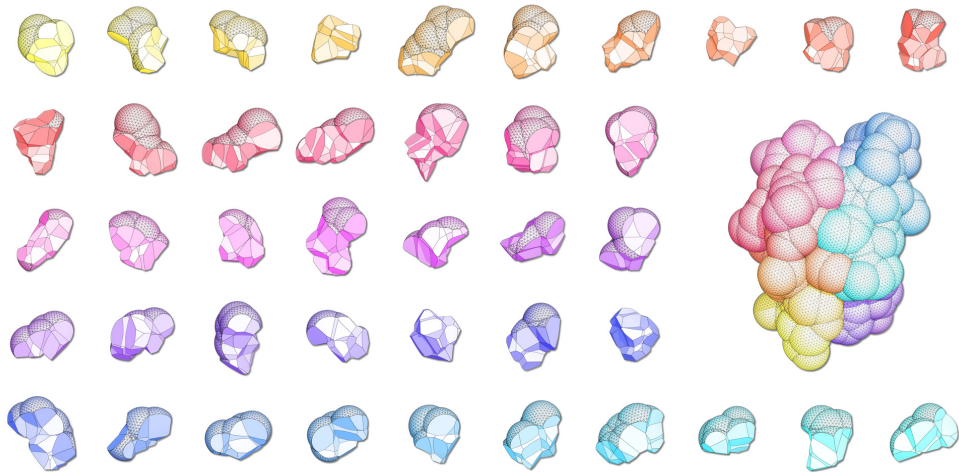
Constrained Voronoi cells
of a single residue



Both interatomic and solvent
contacts of a single residue

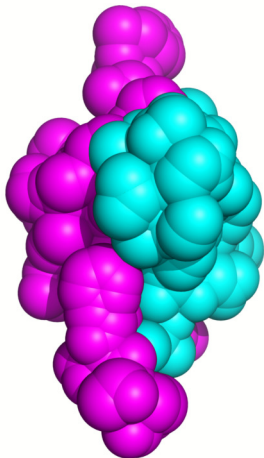


Disassembled tessellation

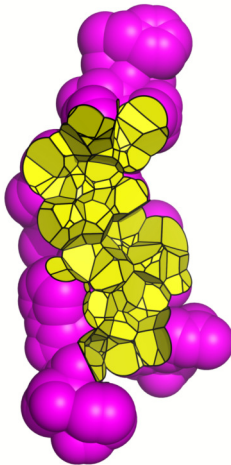


Inter-chain contacts

Solvent-accessible surface
of an insulin heterodimer
PDB:4UNG colored by subunit



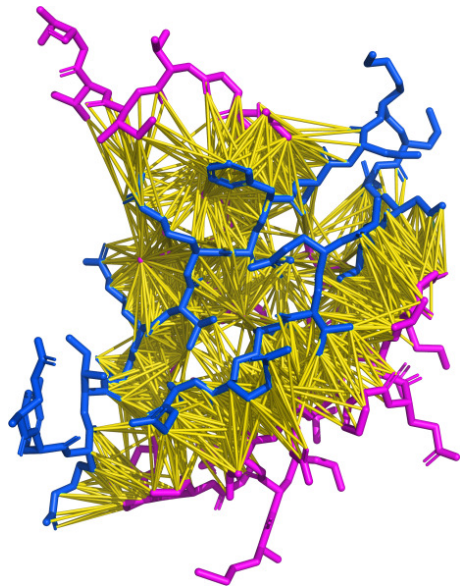
The intersubunit interface
shown together with the
SAS of one subunit



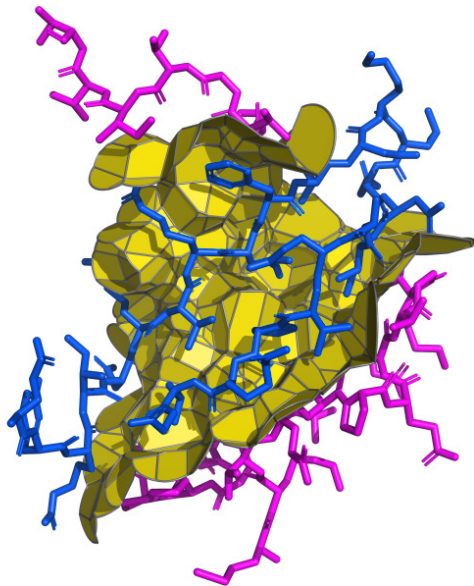
The intersubunit interface
shown together with
both subunits represented
as cartoons



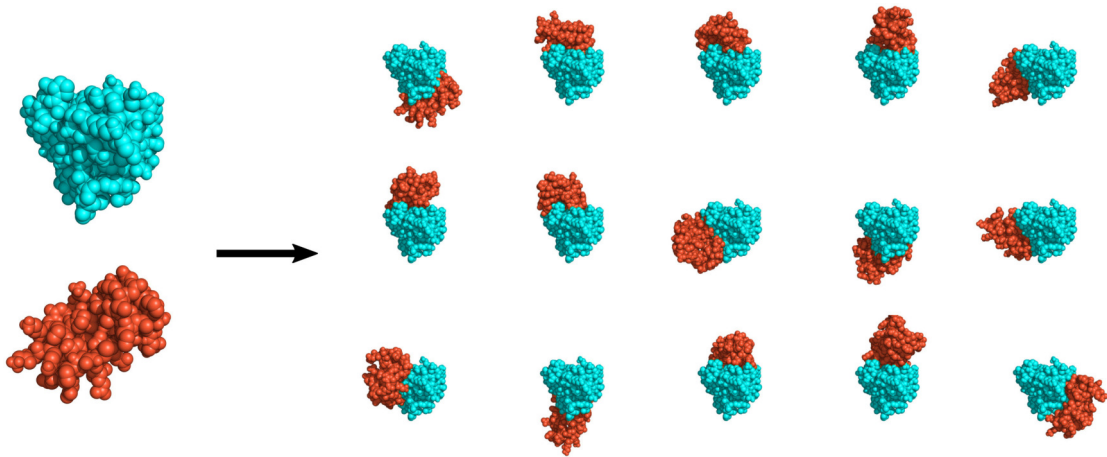
Inter-chain contact areas vs distances



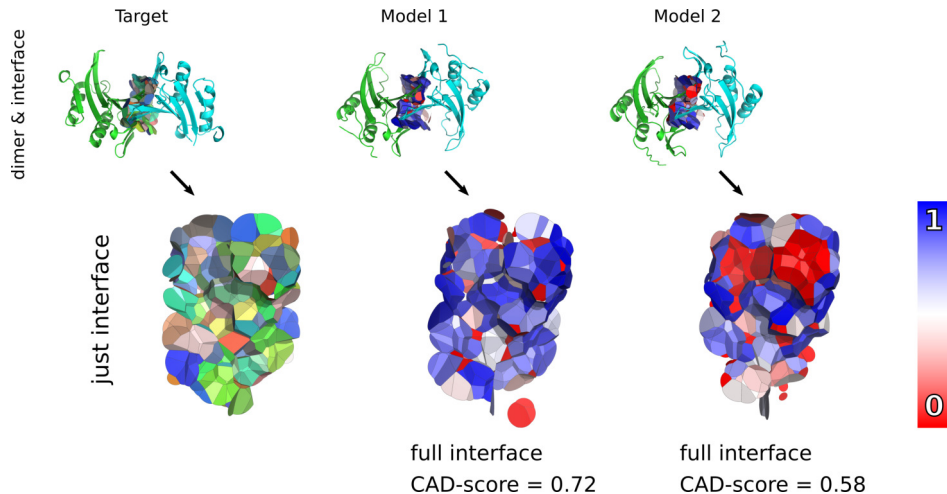
VS



Same chains can have differently modelled interfaces

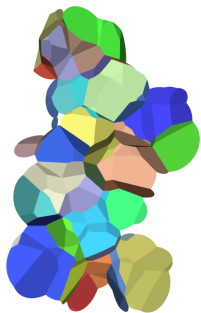


Comparing interfaces using CAD-score (Contact Area Difference score)



Olechnovic and Venclovas. *Contact Area-Based Structural Analysis of Proteins and Their Complexes Using CAD-Score*. Methods in Molecular Biology (2020)

Evaluating interfaces with an area-based potential (e.g. VoronMQA)

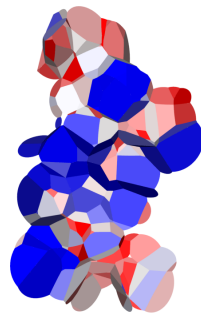


Interface
contact areas



$$E(a_i, a_j, c_k) = \log \frac{P_{\text{exp}}(a_i, a_j, c_k)}{P_{\text{obs}}(a_i, a_j, c_k)} =$$
$$= \log \frac{F_{\text{exp}}(\text{area}(a_i), \text{area}(a_j), \text{area}(c_k))}{F_{\text{obs}}(\text{area}(a_i, a_j, c_k))}$$
$$E_n(\Omega_\phi) = \frac{\sum_{\omega \in \Omega_\phi} E(\text{type}_\omega) \cdot \text{area}_\omega}{\sum_{\omega \in \Omega_\phi} \text{area}_\omega}$$

Statistical potential
for contact areas

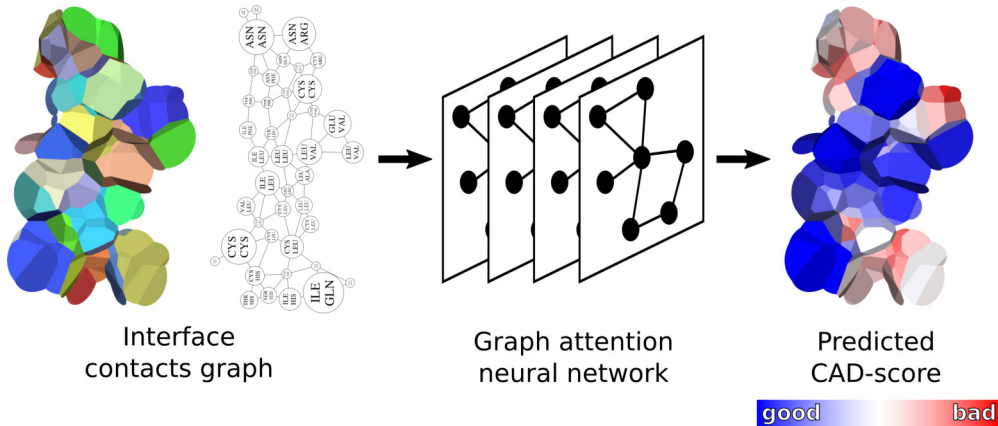


Interface
pseudo-energy



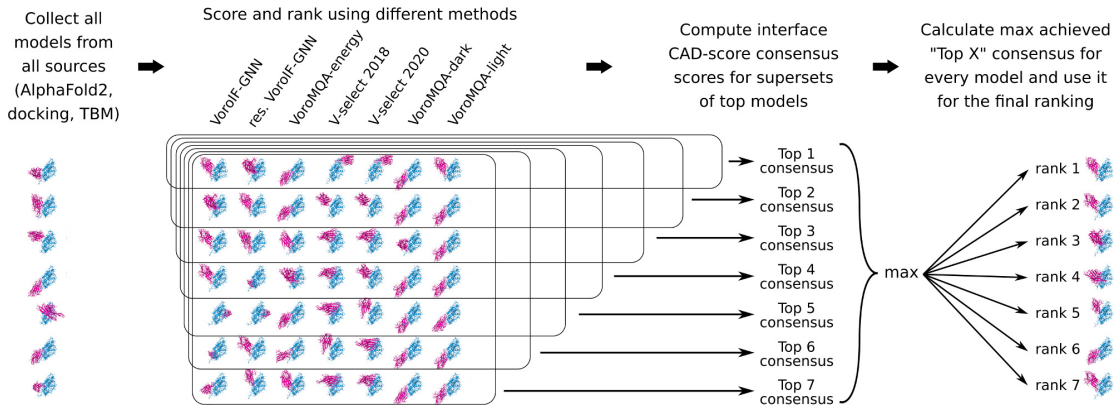
Olechnovic and Venclovas. *VoronMQA: Assessment of protein structure quality using interatomic contact areas*. Proteins (2017)

Evaluating interfaces with a graph neural network (e.g. VorolF-GNN)



Olechnovic and Venclovas. *VorolF-GNN: Voronoi tessellation-derived protein-protein interface assessment using a graph neural network*. Proteins (2023)

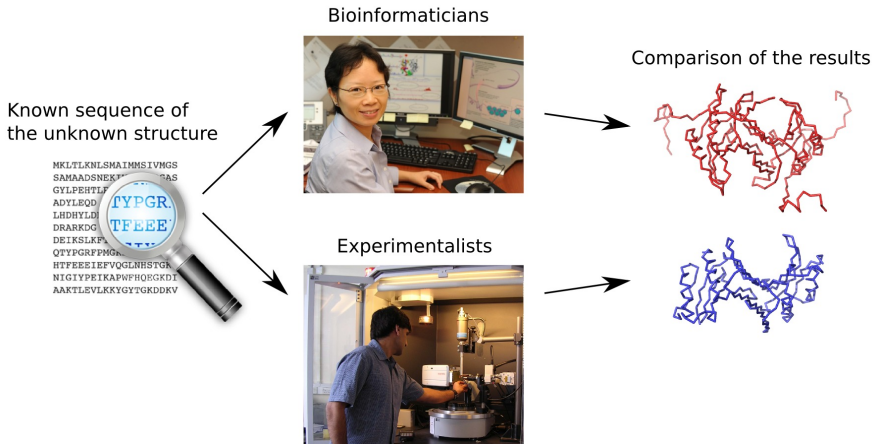
Selecting complex models using VorolF-jury (a.k.a. FTDMP)



VorolF-jury was the best-performing multimeric model selector in 2022 CASP and CAPRI double-blind challenges.

CASP and CAPRI double-blind challenges

The main way to test new methods in structural bioinformatics are the community-wide double-blind testing experiments — CASP and CAPRI.



Our results in CASP and CAPRI challenges over the years

CASP (Critical Assessment of Techniques for Protein Structure Prediction) and CAPRI (Critical Assessment of PRedicted Interactions) are world-wide experiments focused on the blind testing of methods for protein structural bioinformatics.

After
AF2

Before
AF2

- | | |
|------|---|
| 2024 | Top performance (ranked 1st in the CAPRI assessment of scoring groups) in the CASP16-CAPRI scoring experiment. Group “Olechnovic”. |
| 2024 | Contributed to the top performance (ranked 1st) in modeling structures of protein complexes in the CAPRI experiment rounds 47–55. Group “Venclovas”, members: Dapkūnas J, Olechnovič K, Venclovas Č. |
| 2022 | Contributed to one of the top performances in modeling structures of protein complexes in CASP15 and CASP15-CAPRI experiments (ranked 2nd in CASP, ranked 1st in CAPRI). Group “Venclovas”, members: Olechnovič K, Valančauskas L, Dapkūnas J, Venclovas Č. |
| 2022 | One of the top performances in EMA (estimation of model accuracy) in CASP15 experiment (and ranked 1st in CASP15-CAPRI scoring experiment). Groups “Venclovas” and “VoroIF”. |
| 2020 | Contributed to one of the top performances in modeling structures of protein complexes in CASP14 and CASP14-CAPRI experiments (ranked 2nd in CASP, ranked 1st in CAPRI jointly with two other groups). Group “Venclovas”, members: Olechnovič K, Dapkūnas J, Venclovas Č. |
| 2019 | Contributed to one of the top performances (ranked 3rd) in modeling structures of protein complexes in the CAPRI experiment rounds 38–45. Group “Venclovas”, members: Dapkūnas J, Kairys V, Olechnovič K, Venclovas Č. |
| 2018 | Contributed to the best results (ranked 1st) in modeling structures of protein complexes in CASP13 and CASP13-CAPRI experiments. Group “Venclovas”, members: Dapkūnas J, Olechnovič K, Venclovas Č. |
| 2018 | One of the top performances in EMA (estimation of model accuracy) in CASP13 experiment (ranked 1st in prediction of unreliable regions). Groups “VoroMQA-A” and “VoroMQA-B”. |
| 2016 | Contributed to the best results (ranked 1st) in modeling structures of protein complexes in CASP12-CAPRI experiment. Group “Venclovas”, members: Dapkūnas J, Olechnovič K, Venclovas Č. |
| 2016 | One of the top performances in protein structure prediction in CASP12 experiment. Group “VoroMQA-select”. |



CASP15 Team

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- ▶ Voronoi-tessellation-based contact areas can be really useful for the analysis of protein interfaces.
- ▶ Atom-level contact areas can be summed/combined into residue- and chain-level descriptors.
- ▶ Voronoi-tessellation-based contact areas can be constructed and summarized using the **Voronota** software.
- ▶ Tessellation-based description of ground truth antibody-antigen interfaces is proposed for **SEPIQ**.

Thank you for your attention and for inviting me.